

task-1

Simulating a DNS Resolution Failure on Windows

Objective: Simulate a local DNS resolution failure by editing the Windows Hosts file so that **www.spotify.com** points to an incorrect IP address, then test the result with ping.

Open the Hosts file

Open Notepad as Administrator and open:
C:\Windows\System32\drivers\etc\hosts

Add a wrong IP address

Add this line at the bottom of the Hosts file:
192.0.2.1 www.spotify.com Save the file. The 192.0.2.0/24 range is reserved for documentation/testing.

Clear the DNS cache

Open Command Prompt as Administrator and run: ipconfig /flushdns

Expected message:
Windows IP Configuration Successfully flushed the DNS Resolver Cache.

Ping Spotify

ping www.spotify.com
Example result:

```
Pinging www.spotify.com [192.0.2.1] with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Ping statistics for 192.0.2.1:    Packets: Sent = 4, Received = 0, Lost = 4 (100%  
loss)
```

Verify the mapping

```
Run: nslookup www.spotify.com Note: nslookup may query the configured DNS server directly rather than demonstrate the Hosts-file override
```

```
You can also run: ipconfig /displaydns
```

Restore the Hosts file

```
Remove the test line:
```

```
192.0.2.1    www.spotify.com Save the file, then run: ipconfig
```

```
/flushdns
```

Finally test:

```
Ping www.spotify.com
```

Result

The experiment demonstrates that a local Hosts-file entry can override normal name resolution. By mapping www.spotify.com to an incorrect IP address, ping attempts to contact that incorrect address, which can produce Request timed out messages. Removing the entry and flushing the DNS cache restores normal resolution.

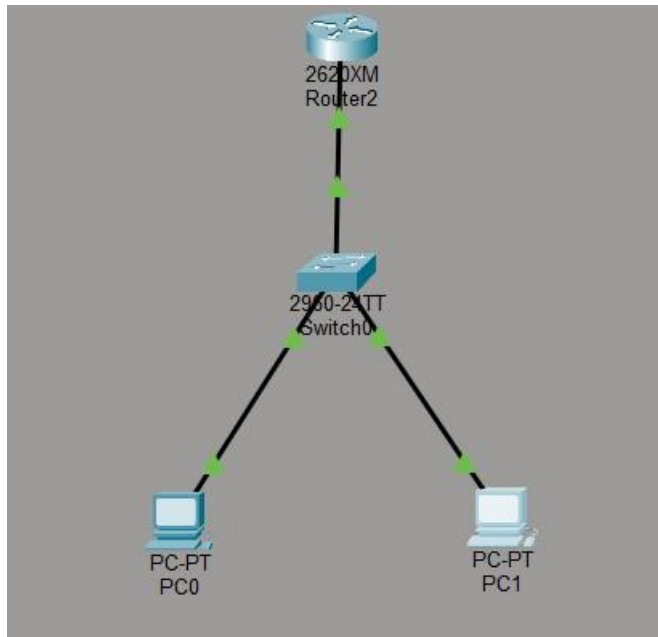
Important: Restore the Hosts file after completing the practical so that Spotify and other services resolve normally.

Task-2

Q-Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

To configure a router as a DHCP server, intentionally assign an incorrect default gateway to clients, verify the assigned gateway, identify the problem, and correct the configuration.

TOPOLOGY:



Router Configuration

enable

configure terminal

interface fa0/0

ip address 192.168.1.1 255.255.255.0

no shutdown exit

```
ip dhcp excluded-address 192.168.1.1
```

```
ip dhcp pool LAN network 192.168.1.0 255.255.255.0 default-router  
192.168.1.254 dns-server 8.8.8.8 end
```

PC Configuration

1. Open **Desktop** → **IP Configuration**.
2. Select **DHCP**.
3. The PCs receive IP addresses automatically.

Example:

Device	IP Address	Default Gateway
PC0	192.168.1.2	192.168.1.254
PC1	192.168.1.3	192.168.1.254

Verification

Run the following command on **PC0**:

```
ipconfig
```

Output:

IP Address.....192.168.1.2

Subnet Mask.....255.255.255.0

Default Gateway.....192.168.1.254

The default gateway is incorrect because the router's actual IP address is **192.168.1.1**.

Problem Identification

The issue was identified by:

- Checking the client's IP configuration using ipconfig.
- Comparing the assigned default gateway with the router interface IP address.
- Reviewing the DHCP pool configuration on the router.

The DHCP pool contained: default-router 192.168.1.254 which is an incorrect gateway.

Solution

Correct the DHCP configuration:

configure terminal

```
ip dhcp pool LAN no default-router 192.168.1.254
```

```
default-router 192.168.1.1
```

```
end
```

Renew the DHCP lease on both PCs by selecting **DHCP** again in the IP Configuration window.

Run:

ipconfig output:

IP Address.....192.168.1.2

Subnet Mask.....255.255.255.0

Default Gateway.....192.168.1.1

The DHCP server initially assigned an incorrect default gateway (192.168.1.254) to the client PCs. The issue was identified using the ipconfig command and by reviewing the router's DHCP configuration. After changing the default-router option to the correct router IP (192.168.1.1) and renewing the DHCP lease, both PCs received the correct default gateway and the configuration was successfully corrected.

Task -3

DNS Troubleshooting – Name Resolution Problem

Scenario:

Devices on the simulated network can access websites using their IP addresses, but they cannot access them using domain names such as google.com. This indicates that network connectivity is working, but **DNS name resolution is not working**.

Troubleshooting Steps

1. Test connectivity using an IP address

ping 8.8.8.8

- If the ping succeeds, the device has Internet/network connectivity.
- If it fails, check the IP address, default gateway, and routing first.

2. Test the domain name

ping google.com

- If 8.8.8.8 works but google.com fails, the problem is most likely DNS.

3. Use nslookup to diagnose DNS

nslookup google.com

A working result should show the DNS server being used and an IP address for google.com.

4. Check the configured DNS server

On Windows:

ipconfig /all

Look for the **DNS Servers** entry.

5. Test a known DNS server

nslookup google.com 8.8.8.8

- If this works, the configured DNS server is probably incorrect, unreachable, or not responding.
- If it also fails, check the network connection or DNS service.

6. Fix the DNS configuration

Configure the client with a valid DNS server, for example:

7. Preferred DNS: 8.8.8.8

Alternate DNS: 1.1.1.1

8. Renew the network configuration

9. `ipconfig /release`

10. `ipconfig /renew`

`ipconfig /flushdns`

11. Verify the solution

12. `nslookup google.com`

`ping google.com`

If the domain resolves to an IP address and the ping succeeds, DNS is working correctly.

Conclusion

The problem was caused by an **incorrect or unavailable DNS server**. Since devices could communicate using IP addresses, the basic network connection was working. Using nslookup helped confirm that domain-name resolution was failing. Correcting the DNS server address and flushing the DNS cache resolved the problem.

Task-4

DHCP Troubleshooting – APIPA Address Problem

Scenario:

A device in a simulated network is not receiving an IP address from the DHCP server. Instead, it automatically assigns itself an **APIPA address in the 169.254.x.x range**. This usually means the device could not communicate with the DHCP server.

Possible Causes

1. **Network connectivity problem** – Ethernet cable, switch port, or Wi-Fi connection may be incorrect.
2. **DHCP server is unavailable** – The DHCP server may be powered off or its service may not be running.
3. **DHCP scope is exhausted** – All available IP addresses in the DHCP pool may already be assigned.
4. **Incorrect DHCP configuration** – Wrong network address, subnet mask, default gateway, or DHCP pool settings.
5. **Incorrect VLAN configuration** – The client and DHCP server may be placed in different VLANs without proper routing.
6. **DHCP relay problem** – If the server is on another network, the router may not have a correct DHCP relay (ip helper-address).
7. **Client is configured incorrectly** – The device may be using a static IP or DHCP may be disabled.

Step-by-Step Troubleshooting

Step 1: Check the IP address

On Windows, run:

```
ipconfig
```

If the device shows something like:

IPv4 Address: 169.254.10.25

it indicates that the DHCP request was unsuccessful.

Step 2: Check physical/network connectivity

- Check that the cable is connected.
- Verify the switch port is active.
- Check that the correct network interface is enabled.
- In Packet Tracer, verify that the link indicators are green.

Step 3: Check the DHCP server

Verify that:

- The DHCP server is powered on.
- DHCP service is enabled.

- A valid DHCP pool/scope exists.
- The pool has unused IP addresses.

For example:

Network: 192.168.1.0/24

Gateway: 192.168.1.1

DHCP Range: 192.168.1.100 – 192.168.1.200

DNS: 8.8.8.8

Step 4: Check DHCP scope exhaustion

If all addresses from 192.168.1.100 to 192.168.1.200 are already assigned, increase the DHCP pool or remove unused leases.

Step 5: Check router/DHCP relay

If the DHCP server is on a different subnet, check the router interface:

show running-config

Make sure the appropriate interface contains a DHCP relay such as:

ip helper-address 192.168.1.10

where 192.168.1.10 is the DHCP server's IP address.

Step 6: Renew the client's IP address

On Windows:

```
ipconfig /release
```

```
ipconfig /renew
```

Then check:

```
ipconfig
```

The device should now receive an address from the correct DHCP range instead of 169.254.x.x.

Step 7: Test connectivity

Test the default gateway:

```
ping 192.168.1.1
```

Then test the DHCP server:

```
ping 192.168.1.10
```

Finally, test Internet connectivity if applicable:

```
ping 8.8.8.8
```

Expected Result

Before troubleshooting:

IP Address: 169.254.x.x

DHCP: Failed

After fixing DHCP:

IP Address: 192.168.1.100

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

Conclusion

An **APIPA 169.254.x.x address** indicates that the client failed to obtain an address from DHCP. The main checks are **network connectivity, DHCP server availability, DHCP scope exhaustion, and DHCP/relay configuration**. After correcting the problem, renewing the DHCP lease should provide the client with a valid IP address and restore network connectivity.

Task-5

Type	Issue	Symptoms	Possible Causes	Recommended Solution
DNS	DNS Server Unavailable	Websites cannot be accessed by name, but IP addresses work	DNS server is down, unreachable, or incorrectly configured	Check DNS server availability and configure a valid DNS server such as 8.8.8.8

Type	Issue	Symptoms	Possible Causes	Recommended Solution
DNS	Incorrect DNS Configuration	nslookup fails or returns “DNS request timed out”	Wrong DNS server IP configured on the client	Check with ipconfig /all and correct the DNS server address
DNS	DNS Cache Problem	Old or incorrect IP address is returned for a website	Corrupted or outdated DNS cache	Run ipconfig /flushdns and try the lookup again
DHCP	DHCP Server Unavailable	Client receives a 169.254.x.x APIPA address	DHCP server is offline or DHCP service is stopped	Start the DHCP service/server and verify network connectivity
DHCP	DHCP Scope Exhausted	New devices cannot obtain an IP address	All addresses in the DHCP pool have been allocated	Increase the DHCP scope or remove unused DHCP leases
DHCP	Incorrect DHCP Configuration	Client receives an incorrect IP, gateway, or DNS address	Incorrect DHCP pool, subnet mask, gateway, or DNS settings	Correct the DHCP scope settings and renew the client's lease using ipconfig /renew